

ALAN VERAS

AI Developer | LLM, RAG & Agentic Systems in Production | Python · LangGraph · AWS · Docker

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PROFESSIONAL SUMMARY

I build and operate **LLM systems in production**, and I am the **AI engineer** who also owns the **infrastructure** underneath them. At **Youlex** (legaltech), I shipped **60+ production improvements in 6 months**, covering **LLM cost observability**, latency reduction, **LLM provider resilience**, and **IAM hardening**, and I operated the **8-stage data/ML pipeline** with **automated quality gates**. As an independent consultant, I provisioned and operated infrastructure on **VPS** (DigitalOcean, Contabo, Hetzner) running **containerized AI agents 24/7**, and delivered **computer vision systems** for retail and agribusiness clients. On the AI side, I orchestrate **legal agents in LangGraph** and have designed **77+ agentic LLM evaluation tasks**, spanning adversarial prompt evaluation and tool-using agents.

TECHNICAL SKILLS

Infrastructure, Cloud & DevOps: Linux (administration and provisioning on VPS: DigitalOcean, Contabo, Hetzner), AWS (EC2, ECR, ECS, S3, CloudWatch, IAM, VPC/Security Groups), Docker, CI/CD (GitHub Actions), Cloudflare (edge, R2), Nginx/reverse proxy, Networking and DNS, Secrets Management, Shell Scripting/Bash

Observability & Reliability: Amazon CloudWatch (metrics, logs, and alarms), Grafana, automated quality gates, latency monitoring (p95), SLA and resolution-rate metrics

Automation & Data Platform: Pipeline orchestration (LangGraph, ARQ/Redis), n8n, artifact versioning, PostgreSQL, Supabase, Redis, REST APIs, WebSockets

AI & RAG Systems: Hybrid RAG (BM25 + dense + RRF fusion), Rerankers (BGE), Legal-BERTimbau, PyTorch, Transformers (HuggingFace), BERTopic, Computer Vision (YOLO, OpenCV, SAHI, dataset annotation), LLM Integration (Gemini, GPT-4, Claude, DeepSeek), LLM Evaluation (77+ agentic tasks), Model Serving (FastAPI, SSE streaming)

Languages & Frameworks: Python, FastAPI, Bash, TypeScript, Node.js (Nest.js, Express.js, Fastify), Go, React, Next.js

Applied Security: PII detection and sanitization, guardrails and safety gates, RBAC/ABAC, audit logging and traceability, authentication hardening, prompt injection testing

Practices: pytest, Jest, Vitest, TDD, Agile/Scrum, Technical Leadership, Multi-Agent Workflows (8+ agents in roundtable code-review loops)

PROFESSIONAL EXPERIENCE

Youlex

March 2026 - August 2026

Founding Engineer: Infrastructure & AI / Legaltech

- Orchestrated **multi-provider AI agent teams** (Claude, GLM, and others) with **human review** and **per-stage quality gates**, shipping **60+ production improvements in 6 months**
- Built and operated the **8-stage data/ML pipeline** processing **7.2 GB** of TCU open data (**199k case rulings**, 17,883 curated docs): ingestion, **zero-PII sanitization**, topic modeling (**BERTopic**), and embeddings, with **automated per-stage quality gates** and versioned artifacts published only after passing a **Recall@5 $\geq$ 90%** gate
- Drove the **infrastructure cost-down** by resizing **EC2 instance sizes** based on real **CloudWatch** metrics (peak RAM measured before each provisioning)
- Drove the **migration of the LangGraph agent chain** (reviewer, auditor, researcher) from **individual provider APIs** to **OpenRouter**, cutting **cost** and raising **answer quality** with per-tier model selection
- Built the **LLM resilience layer** on OpenRouter: provider abstraction with **circuit breaker**, **retry**, and **context cache**; **cascade fallback** validated by **A/B testing** before go-live; native **prompt caching** (Anthropic ephemeral + DeepSeek)

- Formalized **CI/CD for database migrations** on **GitHub Actions**; **diagnosed and fixed in production** a deploy embedding `localhost:8000` that took the frontend down; performed **IAM hardening on AWS (least-privilege policy)** for the **ECS** control plane validated with **simulate-principal-policy**; 30-day retention on 5 containers' logs)

Self-employed

February 2025 - April 2026 (transition to Youlex)

Independent Consultant: Infrastructure & AI Engineering

- Provisioned and operated **24/7 infrastructure on Linux VPS across multiple providers** (DigitalOcean, Contabo, Hetzner), sizing instances and administering **Docker**, **n8n**, Postgres/Supabase, and **Redis** for AI products, with **WhatsApp API integration** (Uazapi/Evolution) into **Gemini + OpenAI** and SLA, resolution, and funnel metrics instrumentation
- Delivered end-to-end **WhatsApp AI agents** for **Arte Studio** (Aug to Dec operation): response time from **6 to 12h down to 1 to 3min, 24/7**, with automated triage, lead qualification, and follow-up across **10 to 15 conversations/day** and **human handoff rules** for sensitive cases
- Developed a **computer vision system** for **Onflux** to **detect cars and people** in commercial environments, using **YOLO** and **OpenCV**, tested on **fish-eye camera datasets**, shopping mall cameras, and web datasets
- Worked directly with **agribusiness players** on **computer vision**: captured and annotated **1,000+ fruit images** and, with my team, trained and benchmarked **YOLOv8** and **YOLOv11** with techniques such as **SAHI** and augmentation, developing a **custom model** that reached **~70% accuracy on real-world images**
- Cut feature delivery time by **3x (2 to 3 days vs. 1 to 2 weeks)** with a **multi-agent CI/CD workflow** of **8+ specialized AI agents** (code auditor, test generator, impact reviewer)

Projeto Ares

November 2024 - Present

General Directorate of Extension and Research: Ares Project (UFC)

- Progressed from member (**digital forensics studies**) to **blue team researcher** and, now, to the **General Directorate of Extension and Research**, leading a team of **~14 members across 3 squads** (Communication, Development, Research) through the project's 2025 institutional pause and 2026 relaunch
- During this period, the project delivered: **acceptance into a selective program at Instituto Atlântico** (2024), a **published paper** (2025), **minicourses** in 2025 and at UFC (2026), and **lectures** with a digital forensics expert and a blue team professional
- Followed the development of the **Deimos, Argos, and Cerberus projects: cybersecurity and AI tools for network analysis**, built in **C and Python**, inspired by well-known **pentesting** technologies such as **Nmap** and **Wireshark**
- Implemented a **weekly delivery cadence** (weekly, planning, retrospectives) and a **Kanban** backlog, sustaining continuous delivery of extension projects, documentation, and events
- Co-built **training materials and documentation** for the technical ramp-up and onboarding of new members, and organized **1 in-person security CTF** with **~15 to 20 participants**

Outlier

June 2026 - Present

LLM Evaluation: Agentic Tasks (Freelance)

- Designed **77+ evaluation tasks** for agentic LLMs, spanning **adversarial prompt evaluation** and prompt security (60+) and tool-using agent behavior (17+), and author **weighted rubrics** across Task Completion, Instruction Following, Factuality, Tool Use, and Safety, with atomic criteria anchored in the prompt
- Run the model in-platform and **score its trajectories by hand** against the rubric with evidence-anchored justifications, identifying **genuine model failure modes** and discarding levers that do not discriminate

EDUCATION, CERTIFICATIONS & LANGUAGES

Computer Science (Artificial Intelligence emphasis) | Federal University of Ceará | 2023 - 2027 | **Languages:** English (Advanced), Portuguese (Native)

Certifications: AWS Cloud Practitioner and CKA (in preparation) | SQL for Data Science